Research Consent Form

MIAMI UNIVERSITY
OXFORD, OH • EST. 1809

Dear participant:

Thank you for considering participation in this study. I am Dr. Jay Shan, an Associate Professor in the Department of Information Systems & Analytics at Miami University. My research team and I are conducting a research project to develop an ecosystem that integrates web-based virtual reality (VR) and augmented reality (AR) for personalized safety training and the goal of this study is to evaluate user feedback of this system.

If you agree to participate, you will be asked to complete 2 sessions, separated by approximately 1 week. Each session will last approximately 2 hours. You will receive a $75 Visa gift card at the end of each session you attend. At the start of the first session, we will ask you to complete a demographic questionnaire and two other questionnaires on hazard evaluation and technology. This should take approximately 20 minutes. We will then ask you to complete one of two training conditions depending on the session. The two conditions are training either with a web-based VR system or with an AR headset system. The VR session will involve a web-based VR scenario presented on a standard 27" monitor where you will be seated and interact with a standard mouse and keyboard to train on a lathe, milling machine, and collaborative robot. For the AR session, you will wear a head-mounted display to complete first person perspective training while standing at a non-operational milling machine, a lathe that runs behind its closed door, and a collaborative robot that moves slowly and stops if it touches anything. The headset shows a zone near the robot that you must not enter.

At the start of the training portion of the session, you will receive a system orientation, where you will learn about using the system. Then, we will ask you to complete three training scenarios. Each of the scenarios will be focused on recognizing and mitigating manufacturing workplace hazards and machine operation. Between scenarios, you will be able to take a break. Following completion of the three scenarios, we will ask you to complete one assessment module per machine (VR only), a post-session feedback questionnaire, and an interview. Your performance on the training scenarios will be recorded.

You will not be asked to include your name on any of the questionnaires, thus your answers cannot be associated with you. The questionnaire responses will be treated as confidential information, stored in a secure location for at least three years after the study, and accessed only by the research team members, under the supervision of Dr. Fadel M. Megahed.

Your questionnaires will be scanned into a password-protected study website. OpenAI, an artificial intelligence service, will help read the handwriting and find themes in the interview answers after names are removed, and a research team member will check every response. With your permission, the interview (approximately 5–10 minutes, 4 questions) will also be recorded and transcribed offline on the SIGHT workstation; otherwise, the researcher will take notes. Recordings will be labeled with your study ID only and deleted once the transcript is checked. According to OpenAI's documentation (https://developers.openai.com/api/docs/guides/your-data), information sent through its API is not used to train its models.

Your participation is voluntary and you may withdraw from the study at any time or decline to answer any questions that make you uncomfortable. If you withdraw, your responses will be retained unless

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you ask that they be deleted. You will not be asked to do anything that exposes you to risks beyond those of everyday life. During the study, if you feel any discomfort with the use of the VR or AR systems, you may ask to stop the session. The benefit of the study, scientifically, is to help develop a proof-of-concept training system that integrates AI-driven safety insights, gamified web-based VR, and AI-powered AR for real-time safety and productivity guidance. This project aims to provide an immersive, adaptable, and scalable hazard training tool for Ohio's manufacturing workforce. At the end of the study, the researcher will make the general results available to you. The generalized results may be presented at professional conferences or published in articles describing the results of the research. Only aggregate results and unidentified interview quotes will be reported.

When using virtual reality systems, you could experience one or more of the following: disorientation, dizziness, eye strain, headache, lightheadedness, nausea, or sweating. It is possible that these could linger after participation, or that there may be unknown aftereffects not identified by current research. If there is any change in the way you are feeling, please let the researchers know immediately. If you feel any effects, you will be asked to remain at the lab until any side effects have subsided.

If you choose to participate in the study, you may discontinue participation at any time without penalty or loss of benefits. By signing the form, you do not give up any personal legal rights you may have as a participant in this study. You may refuse to participate in this study without penalty of loss of benefits to which you are otherwise entitled.

If you suffer an injury from participation in the study, you should notify the researcher immediately. By signing this form you acknowledge and agree that (1) the cost for medical treatment will be billed to you or your insurance, and (2) Miami University will not be responsible for any medical bills incurred while participating in this study. Miami University has no funds set aside for the payment of health care expenses for this study.

If you have any questions about the research now or in the future you may contact me at jayshan@miamioh.edu or (513) 529-2135. If you have any questions or concerns about the rights of research subjects or the voluntariness of participation, you may contact the Research Ethics and Integrity Program at Miami University at either (513) 529-3600 or humansubjects@miamioh.edu.

Thank you for your participation. We are very grateful for your help and hope that this will be an interesting session for you. You may keep this portion of the page. This project has been reviewed and approved by Miami Institutional Review Board (Level 2): Protocol Number 02415r.

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I agree to participate in this study about VR and AR training for workplace safety. I understand my participation is voluntary and that my name will not be associated with my responses. By signing below, I acknowledge that I am 18 years or older.

Audio recording of my interview (about 5–10 minutes, 4 questions): ☐ I agree ☐ I do not agree

Participant Signature _____ Date: _____

Researcher Signature _____ Date: _____

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